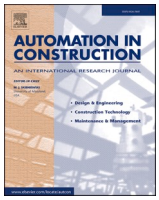
Automation in Construction 179 (2025) 106482



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





# Automated knowledge graph-based risk assessment for fall-from-height accidents in construction

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ARTICLE INFO

*Keywords:*
Fall-from-height
Accident reports
Knowledge graph
Large language models
Risk factor analysis

ABSTRACT

Fall-from-height (FFH) accidents remain a leading cause of fatalities in the construction industry. To systematically extract and analyze risk factors from unstructured FFH accident reports, this paper employed large language models (LLMs) to enable zero-shot automated fall-from-height knowledge graph (FFHKG) construction. By clustering FFHKG entities to merge semantically similar factors, a weighted complex network is formed, enabling topological analysis for quantitative risk assessment. A case study on 1097 FFH accident reports validates the proposed framework. Results demonstrate that GPT-4o achieves high extraction accuracy, with an F1 score of 0.94 in named entity recognition and a precision of 0.90 in relationship extraction. Key risk factors, such as poor safety management, lack of training, insufficient edge protection, etc., are quantitatively identified across multiple perspectives. Observable unsafe behaviors are also detected, offering insights for behavior-based safety monitoring. The proposed framework provides a data-driven solution for more effective safety management on construction sites.

## 1. Introduction

The construction sector serves as a fundamental component of the national economy. Nevertheless, the combination of mechanized operations, manual labor, and the inherently complex and ever-changing work environment poses persistent safety challenges. Among all accident types in this field, fall-from-height (FFH) accidents remain the primary cause of fatalities, contributing to roughly 40 % of construction-related deaths [1]. For example, in 2020, there were 4764 recorded fatalities in the U.S. construction industry [2], while the UK reported 39 fatal cases (equivalent to a rate of 1.62 deaths per 100,000 workers), markedly higher than the 0.65 per 100,000 observed in the manufacturing sector [3]. Although advancements in safety management have led to a gradual decline in FFH-related fatalities, these accidents remain a pressing concern worldwide. Effectively reducing FFH accidents requires a thorough understanding of their underlying risk factors and accurate risk assessment methods to support decision-making in construction safety management.

To evaluate FFH risk factors, researchers have traditionally relied on expert knowledge to construct expert-based systems that quantify the importance of various factors [4,5]. However, expert judgment is inherently subjective, and obtaining reliable and extensive expert ratings remains challenging [6]. The construction industry has accumulated a vast number of accident reports that are now readily accessible online. Leveraging historical accident data offers a more objective approach to identifying risk factors and trends, reducing the potential bias introduced by expert assessments [7]. Existing data-driven methods for accident analysis can be broadly classified into two categories. The first approach involves statistical analysis of structured accident data, where researchers extract patterns related to accident frequency, severity, and distribution [8,9]. While this method effectively uncovers historical trends, it struggles to capture complex causal interactions among risk factors. Construction accidents often result from a chain reaction of multiple hazards rather than isolated causes. To address these limitations, researchers have introduced System Accident Analysis (SAA) models to analyze accident mechanisms [10]. Classical models such as the Domino Model, Swiss Cheese Model, Accident Mapping (AcciMap), System-Theoretic Accident Model and Processes (STAMP), and Functional Resonance Analysis Method (FRAM) attempt to represent accident causality [11]. These models offer valuable insights into how risk factors interact and escalate into accidents. However, they primarily serve as conceptual frameworks and lack the capability to

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https://doi.org/10.1016/j.autcon.2025.106482
Received 6 May 2025; Received in revised form 13 August 2025; Accepted 15 August 2025
Available online 28 August 2025
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perform quantitative analysis, making it difficult to accurately measure the degree of influence among risk factors.

Network theory provides an effective means of quantitatively analyzing and visualizing complex causal relationships [12]. Among various network-based approaches, knowledge graph (KG) has emerged as a powerful tool for capturing and representing structured knowledge [13]. By abstracting interrelated entities as connected nodes and defining their relationships, KG has been widely applied in knowledge modeling and risk analysis across multiple domains [14,15]. In the context of construction safety, KG offers a promising approach for the multi-dimensional modeling of accident causality. However, a major challenge in KG construction for construction safety lies in the unstructured nature of accident reports. Existing approaches often rely on manual annotation and coding, which is labor-intensive, time-consuming, and impractical, given the increasing volume of accident records [16]. The emergence of Natural Language Processing (NLP) techniques has enabled automated knowledge extraction from unstructured text, paving the way for KG automation. Some studies have trained information extraction models using manually annotated datasets [17]. However, the construction safety domain lacks large-scale, well-annotated corpora, limiting the generalization ability of such models.

With the rapid advancement of NLP, Large Language Models (LLMs) have demonstrated remarkable capabilities in processing and understanding human language, significantly enhancing the efficiency of information extraction [18]. LLMs offer a scalable solution for analyzing vast amounts of unstructured accident reports, enabling the automated construction of KG without relying on extensive labeled datasets. A promising technique in this domain is prompt engineering, which optimizes LLM output for downstream tasks without requiring model retraining [19]. By formulating structured prompts, researchers can guide LLMs to extract relevant causal relationships from accident reports flexibly and resource-efficiently. This approach reduces the dependence on large, annotated datasets while maintaining adaptability across different accident scenarios. Given the increasing complexity of risk factor interactions in FFH accidents, leveraging LLMs for automated KG construction offers a powerful tool for advancing data-driven accident analysis.

This paper aims to leverage LLMs to construct fall-from-height knowledge graphs (FFHKG) from unstructured accident reports to quantitatively analyze risk factors in FFH accidents. The main research questions in this paper are: (1) How can structured knowledge be accurately extracted from unstructured accident reports under a zero-shot learning paradigm to construct a reliable FFHKG? (2) How can the FFHKG be used to quantitatively analyze causal relationships and identify key risk factors for FFH accidents? To solve these questions, this paper first constructs the FFHKG automatically using LLMs. The entities within the FFHKG are subsequently semantically embedded and grouped using hierarchical clustering, enabling the fusion of similar entities and the transformation of the FFHKG into a weighted complex network that is easier to analyze. Topological analysis is then applied to quantify the influence of each risk factor from different perspectives. The novelty of this research lies in (a) developing a KG structure integrated with the AcciMap model, enabling a comprehensive representation of multi-level causality in FFH accidents; (b) constructing a prompt-based information extraction pipeline that enables zero-shot, LLM-powered KG construction without the need for annotated training data; (c) conducting a systematic and objective quantitative analysis of risk factors through the networked causal structure extracted from a large corpus of real-world accident reports.

The remainder of this paper is structured as follows. Section 2 comprehensively reviews existing studies on FFH accident factors analysis and automated KG construction. Section 3 details the proposed methodology. Section 4 reports the results of a case study of 1097 FFH accidents, while Section 5 provides further discussion. Finally, Section 6 concludes the paper and outlines limitations and directions for future research.

## 2. Related studies

Understanding the underlying causes of FFH accidents and effectively structuring the associated knowledge are both critical for improving construction safety. Therefore, this section reviews existing studies on FFH risk factor analysis and the automated construction of knowledge graph from unstructured accident data.

### 2.1. Risk factors analysis of fall-from-height accident

Numerous studies have investigated the risk factors associated with FFH accidents. Based on the existing literature, these factors can be broadly categorized into four groups: organizational and management factors, equipment-related factors, environmental factors, and individual factors. Organizational and management risks include insufficient safety training, poor management practices, and inadequate safety supervision. For example, Luo et al. [20] utilized the Apriori algorithm to examine 557 FFH accidents reported on construction sites from 2013 to 2019 and identified insufficient safety training and site complexity as major contributing factors. Similarly, Nawi et al. [21] used questionnaire data analyzed through a Likert scale and reported that inadequate education and poor management were primary contributors to FFH accidents. Equipment-related factors mainly include the absence or improper use of personal protective equipment (PPE). Kang [22] conducted frequency analysis alongside chi-square and Kruskal–Wallis tests on FFH accident data from 1997 to 2012, revealing that more than 70 % of these accidents were linked to the absence of fall protection equipment. Cheng et al. [23] used data mining methods to investigate construction-related occupational accidents in Taiwan and found that the most commonly reported contributors to FFH accidents included improper or missing PPE, insufficient safety awareness, and neglect of hazard warnings. Environmental factors also play a crucial role in FFH accidents. These include complex site layouts, adverse weather conditions, and deficiencies in safety installations such as guardrails and scaffolding. Bobick [24] examined the Bureau of Labor Statistics from 1992 to 2000 and identified the absence of guardrails as a leading contributor to FFH accidents. Halperin et al. [25] evaluated scaffold safety at nine construction sites using a standardized checklist, highlighting missing planks, inadequate guardrails, and poor access as significant hazards. Individual factors primarily involve workers' behavior, experience, and risk perception. Wong et al. [26] reported that poor safety attitudes and behaviors among workers posed major challenges to construction site safety management. Jahangiri et al. [27] also found that improper use of PPE and workers' neglect of protective measures were common risk behaviors. In a study focused on the Malaysian construction sector, Ayob et al. [28] reported that 46.28 % of FFH accidents were linked to safety regulation violations, inadequate use of PPE, unsafe practices, and ineffective management.

However, FFH risk factors do not exist in isolation but are interrelated and mutually influential. For instance, poor management can lead to the absence of safety equipment, thereby increasing the risk of accidents; similarly, unfavorable site conditions may exacerbate unsafe individual behavior. Although previous studies have identified various risk factors, limited attention has been paid to the interactions among them. Traditional statistical methods are insufficient to capture the dynamic interplay between multiple contributing factors. To overcome the limitations of traditional statistical approaches, some studies have adopted SAA models to examine the interdependence among various factors and uncover the underlying mechanisms of accidents in complex construction site environments. The Human Factors Analysis and Classification System (HFACS) framework categorizes accident causes into four hierarchical levels: unsafe acts, preconditions for unsafe acts, unsafe supervision, and organizational influences [29]. Qi et al. [30] annotated data using a modified HFACS framework and automatically extracted risk factors from FFH accident reports, ultimately identifying 19 risk factors across the four categories. Other SAA models, such as

AcciMap and STAMP, offer a broader perspective by considering multiple layers of influence, including individual behavior, organizational practices, and policy-level decisions. However, their application in the FFH safety domain remains limited. In the broader context of construction site safety, Ma et al. [31] integrated natural language processing (NLP) with AcciMap-based causal modeling to extract risk factors from 159 accident reports. Yin et al. [32] combined the AcciMap framework with the Plan-Do-Check-Action methodology to comprehensively present the systemic levels and propagation paths of hoisting safety risks, identifying 36 risk factors across six risk domains.

While SAA models provide a structured framework to capture the hierarchical and interactive nature of FFH accident causation, most existing studies emphasize qualitative classification rather than quantitative analysis of the relationships among factors. This shortcoming hinders the identification of critical risks and the prioritization of preventive interventions. To address this gap, complex network theory presents a promising approach by modeling risk factors as nodes and their interactions as edges. This enables the application of network metrics (e.g., centrality, clustering coefficient) to evaluate the importance and connectivity of risk factors [33]. Given that KG inherently possesses a network structure, it offers an ideal platform to integrate SAA-based classifications with quantitative causal inference. Although complex network theory has been successfully applied in accident analysis in other domains [12,33], it has not yet been widely implemented in the context of FFH risk analysis. Therefore, this paper proposes a method for constructing an automated FFHKG grounded in the AcciMap model. By applying network analysis techniques to FFHKG, we aim to quantify the strength of causal relationships among risk factors, thereby providing a scalable, data-driven framework for comprehensive FFH risk assessment.

### 2.2. Automated knowledge graph construction

Building a comprehensive and efficient KG requires accurate information extraction to transform unstructured text into structured representations. However, accident reports are typically lengthy, unstructured, and contain complex causal relationships, making high-precision and high-efficiency KG construction particularly challenging. Some studies rely on manual extraction due to limited data availability, but this approach is time-consuming, costly, and highly dependent on subjective interpretation. As accident records grow, manual methods become impractical for large-scale knowledge extraction.

To address these challenges, NLP techniques have been employed for automated KG construction. Traditional KG construction involves named entity recognition (NER) and relation extraction (RE) [34]. Gan et al. [35] utilized NLP to extract textual information from maritime accident reports, enabling structured KG queries to reveal accident patterns. Wang et al. [36] tested convolutional neural network and recurrent neural network models for relation extraction in construction safety regulations, achieving an accuracy of 87.5 %. Chen et al. [13] proposed a novel model based on a pre-trained language model for water conservancy safety management, reaching over 85 % accuracy in text recognition. While these studies demonstrate the potential of NLP for KG construction, their effectiveness is highly dependent on large, well-annotated datasets. These models perform well on training data but often fail to generalize across different document formats and domains. Moreover, accident reports vary in structure, covering diverse entities related to people, behaviors, environments, and organizations. Unlike general NLP tasks (e.g., extracting names and locations), KG construction for accident analysis requires not only “recognition” but also deep “understanding” of text content.

Given the limitations of traditional NLP models, LLMs offer a promising alternative for KG construction. LLMs generate diverse outputs based on prompts, making them highly adaptable for information extraction tasks. Wei et al. [37] demonstrated that state-of-the-art LLMs exhibit excellent zero-shot and few-shot extraction capabilities, reducing dependence on large annotated datasets. Ashok et al. [38] introduced Prompt-NER, which enhances entity recognition by prompting LLMs to generate comprehensive lists of potential entities. In a systematic evaluation, Li et al. [39] assessed GPT-3.5’s performance in open information extraction but noted its limitations in handling complex relationship extraction tasks. Trajanoska et al. [40] proposed an LLM-enhanced KG pipeline that integrates entity and relation extraction, demonstrating improved performance in domain-specific knowledge structuring. These findings suggest that LLMs can enhance KG construction efficiency and accuracy for accident analysis.

Despite LLMs’ potential, their application in the construction industry remains at an early stage, particularly for unstructured accident report analysis. Researchers have yet to systematically evaluate LLMs’ performance in extracting construction-specific knowledge from unstructured texts. Current studies lack a comparative analysis of different LLM architectures for accident-related IE tasks. Therefore, this paper aims to develop an LLM-based prompt engineering approach to efficiently process large-scale unstructured accident reports, transforming them into structured FFHKG components for comprehensive risk assessment.

## 3. Methodology

This study develops an LLM-based framework for automated FFHKG construction and quantitative analysis of risk factors from unstructured accident reports. The overall workflow is illustrated in Fig. 1. First, the KG ontology is established by adjusting the classical AcciMap model to capture the hierarchical and causal structure of construction FFH accidents. Then, a prompt engineering strategy is introduced, combining Prompt Chaining, Chain of Thought reasoning, and Self-Consistency sampling to enable zero-shot information extraction for FFHKG generation using LLMs. The entities of FFHKG are embedded into dense vectors and clustered using hierarchical agglomerative clustering to fuse semantically similar nodes. The FFHKG is then transformed into a weighted complex network that is more suitable for structural analysis. Finally, topological analysis is applied to quantify the influence of each factor from three perspectives: direct causality, indirect causality, and accident risk propagation, providing a comprehensive understanding of the key contributors to FFH accidents.

### 3.1. Construction of FFHKG ontology

KG is a structured representation of knowledge in the form of a semantic network, where entities, concepts, and events are linked through meaningful relationships [41]. By embedding semantic metadata into these connections, KG enables data integration, contextual understanding, and advanced reasoning across heterogeneous sources. Ontology provides the formal semantic foundation of KG, defining the types of entities, their attributes, and relationships. As the schema of the graph, ontology ensures consistent interpretation and semantic interoperability across domains.

In the domain of construction safety, KGs offer a promising tool to systematically capture complex causality of accidents. Accident reports often contain detailed but unstructured descriptions of unsafe behaviors, faulty equipment, and organizational failures. To support automated knowledge extraction and causal reasoning in this context, a domain-specific ontology is needed to represent the hierarchical structure and interactions among multiple facilities, personnel, and management layers. To achieve this, the AcciMap framework is adopted as the foundation for constructing the FFHKG ontology. Originally proposed by Rasmussen, AcciMap is designed to identify failures, decisions, and actions across different levels of a socio-technical system and to analyze the interactions among these elements that lead to accidents [42]. As illustrated in Fig. 2(a), the original AcciMap framework comprises six hierarchical levels aligned with risk management structures. This hierarchical structure allows both bottom-up classification and top-down

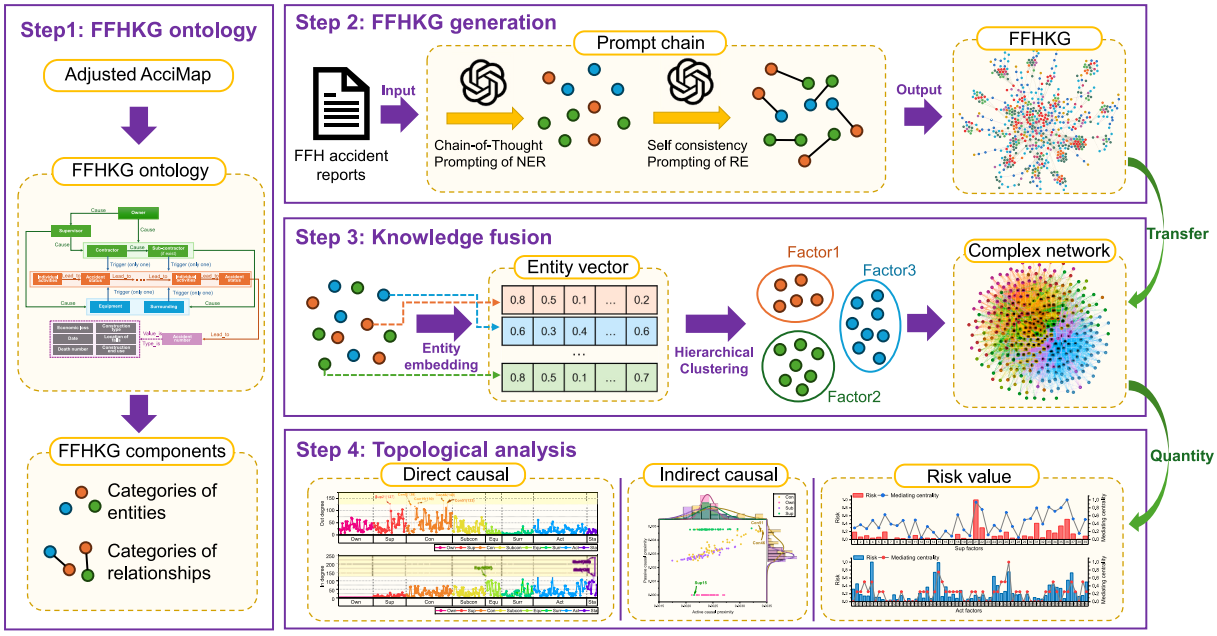



**Fig. 1.** Proposed workflow for automated FFHKG construction and quantitative evaluation.

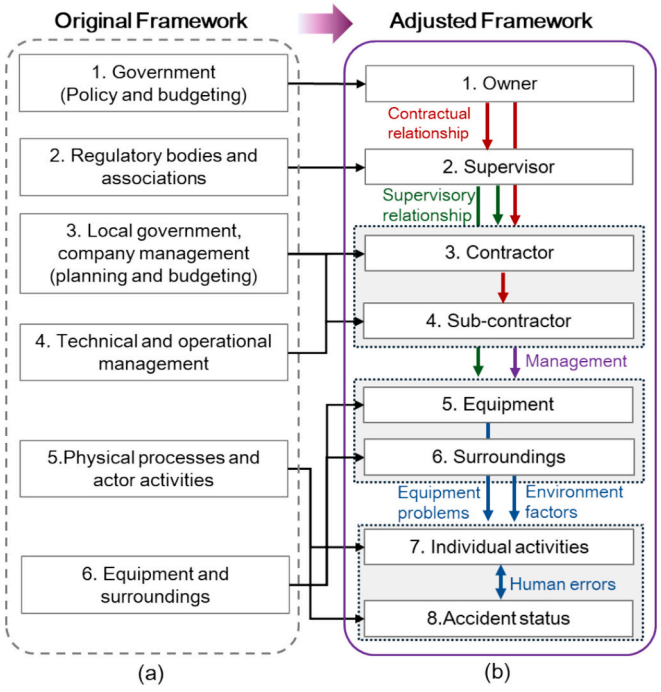



**Fig. 2.** Adjustment details of the AcciMap framework: (a) original framework; (b) adjusted framework.

causality tracing, making it well-suited for modeling accident causation in construction sites [43].

However, the original AcciMap framework lacks the granularity required to capture direct and indirect accident causes comprehensively. To address this limitation, this study modified AcciMap to better suit construction site accident analysis, as shown in Fig. 2(b). Construction sites have a well-established safety supervision system where Owners, Supervisors, Contractors, and Subcontractors operate under contractual and supervisory relationships. Given that these relationships significantly impact safety management, the original AcciMap framework's first four levels have been adjusted to reflect the hierarchical structure of construction projects. The "Government" and "Association" levels in the original framework have been replaced by "Owner" and "Supervisor" to reflect the directly involved entities in construction safety supervision. Additionally, to improve the representation of accident causation, independent "Equipment" and "Surroundings" categories have been introduced as intermediaries between management layers and individual activities. Accidents are influenced not only by worker behavior but also by equipment conditions and environmental factors. Thus, the original fifth level has been divided into Individual Activities and Accident Processes, where worker actions influence accident progression, affecting subsequent behavior. The revised AcciMap framework enhances clarity and consistency in representing causal relationships specific to FFH accidents.

Based on the modified AcciMap framework, this study constructs the FFHKG ontology, as shown in Fig. 3. The ontology consists of eight entity categories corresponding to the adjusted AcciMap levels: Owner, Supervisor, Contractor, Subcontractor, Equipment, Surroundings, Individual activities, and Accident status. In addition to these primary knowledge entities, project-related attributes, such as project type, accident date, location, and consequences, are incorporated to enrich the contextual representation of accidents. A complete list of the knowledge entity types is shown in Table 1.

The relationships between entities are determined based on their type, with six predefined relationship categories. To reflect the multi-layered causality structure of FFH accidents, the ontology defines three types of causal relationships: "Cause", "Trigger", and "Lead_to" [7,14]. Each corresponds to a different stage in the accident formation process. Specifically, "Cause" captures indirect and organizational-level causes; "Trigger" represents direct and immediate causes; and "Lead_to" links the sequential events during the accident itself. By differentiating these stages, the knowledge graph can represent not only who or what contributed to the accident but also how risk factors evolved over time, supporting both root cause tracing and scenario reconstruction. This structure also aligns with layered safety models such as AcciMap, where system-level and individual-level influences are disentangled across time and responsibility dimensions. In addition, two hierarchical relationships are defined: "Value_is" and "Type_is" [44,45]. "Value_is" links entities to quantitative attributes. "Type_is" associates entities with qualitative attributes. This ontology provides a structured and scalable foundation for automated knowledge extraction and risk factor analysis

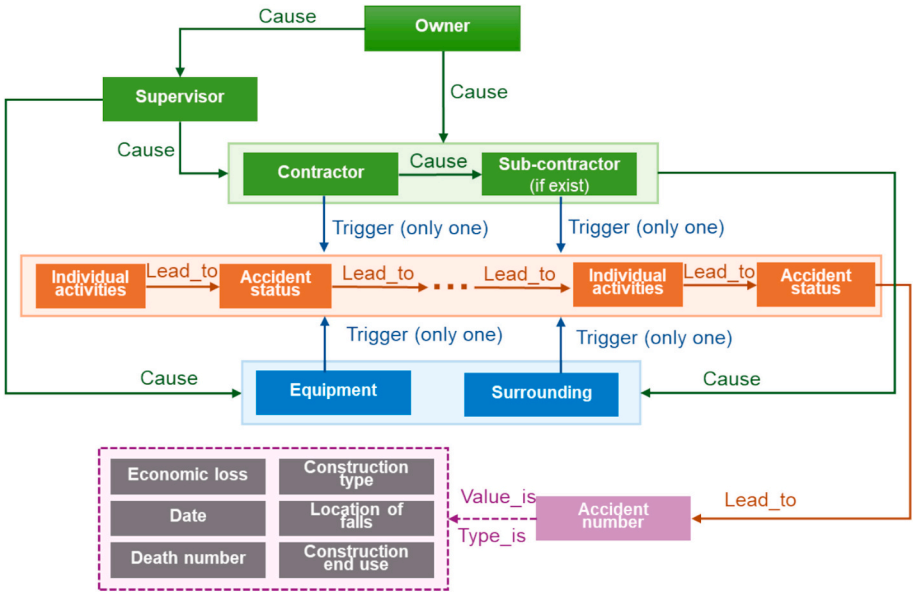



**Fig. 3.** Ontology of the FFHKG.

**Table 1**
Categories of knowledge entities in the FFHKG.

| Entity name | Annotation Tag | Meaning |
|---|---|---|
| Owner | Own | Owner issues related to the accident. |
| Contractor | Con | Contractor issues related to the accidents. |
| Supervisor | Sup | Supervisor issues related to the accidents. |
| Subcontractor | Sub | Subcontractor issues related to the accidents. |
| Surrounding environment | Surr | Environmental issues surrounding the construction site. |
| Equipment | Equ | Problems with worker protection equipment. |
| Individual activity | Act | Individual activity during the accident. |
| Accident status | Sta | Changes in accident status during the accident. |
| Report number | Num | Accident report number in the database. |
| Date | Dat | Date of the accident. |
| Economic loss | Eco | The economic loss of the accident. |
| Death toll | Dea | The death toll of the accident. |
| Location | Loc | Location of the accident. |
| Construction type | Typ | Accident-related construction project type. |
| Construction end use | Use | Accident-related construction project end-use. |

in FFH accidents.

### 3.2. LLM-driven automated generation of FFHKG

NLP serves as the foundation for automatically constructing KG by extracting structured knowledge triples $< e_i, r, e_j >$, where $e_i$ and $e_j$ are entities, and $r$ represents their relationship. With the emergence of LLMs, NLP has unlocked new possibilities for zero-shot information extraction. This section introduces the approach used to construct prompts for LLMs to enable zero-shot extraction of knowledge triples, followed by methods for evaluating extraction quality and constructing and visualizing the resulting FFHKG.

While LLMs have shown the potential to extract complete triples in a single pass directly, their short internal reasoning chains often lead to hallucinated outputs and unstable performance in complex inference tasks. To improve the extraction accuracy of LLMs, this study decomposes the triple extraction task into two classic NLP sub-processes: NER and RE. Then, they are connected through chained prompts (CP), allowing LLMs to focus on a single subtask at a time.

#### 3.2.1. Identification of knowledge entities

NER refers to identifying and classifying words or phrases in text into predefined entity categories. In this study, the information extraction process begins with detecting relevant entities from accident reports. To fully leverage LLMs for the NER task, the prompts are designed based on the following four principles: (1) clear role assignment, (2) contextual learning, (3) Chain of Thought reasoning, and (4) structured output formatting. The complete prompt structure is illustrated in Fig. 4.

First, the LLM is assigned the role of a "construction safety expert." Then, the relevant accident text is provided, and the model is prompted to extract entities based on its contextual understanding. To enhance the reasoning process, Chain of Thought (CoT) reasoning is applied to guide the model through step-by-step logical inference, enabling more accurate and explainable outputs. Since different entity types play distinct roles in accident causation, we designed separate prompt tasks tailored to each entity category. Entities are grouped into three categories. Task A targets management-related entities (e.g., Own, Sup, Con, and Sub), which are typically associated with indirect causes of accidents. Task B focuses on physical context entities, including Equ and Surr, which usually represent direct causes of accidents. Task C captures the timeline chain of the accident by linking Individual Activity with Accident Status. The remaining entity types are basic information and can be extracted through direct queries without requiring CoT reasoning. Finally, the extracted entities are standardized and output in JSON format. This structured output facilitates seamless integration with Python dictionaries, supporting downstream tasks such as relationship extraction between entities.

#### 3.2.2. Extraction of knowledge relation

RE aims to identify relationships between entity pairs in text, essentially classifying the potential relationships between entities. In this study, entity relationships are inferred by prompting LLMs to analyze detailed accident report content. Some relationships, such as "Value_is", "Type_is", and "Lead_to", can be directly determined without complex reasoning. This is because each accident report typically contains only one specific attribute for such fields (e.g., a single economic loss value per accident), making hierarchical relationships like "Value_is" and "Type_is" easily identifiable based on the entity type. Additionally, the "Lead_to" relationship, which connects Individual Activity

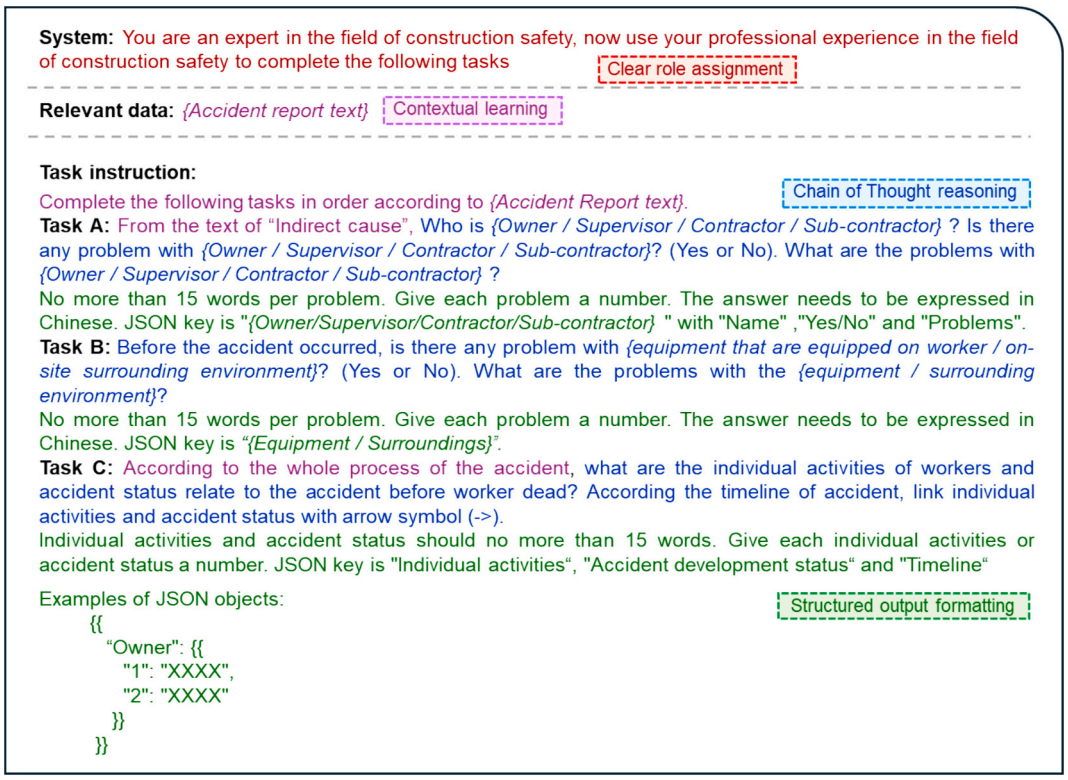



**Fig. 4.** Prompt structure of the NER task.

and Accident Status, can be directly derived from the timeline structure obtained in the previous step. However, the causal relationships of "Cause" and "Trigger" require more intricate reasoning.

To reduce the cognitive burden on LLMs, this study formulates targeted prompts to extract these relationships. For the "Cause" relationship, a question set A is generated based on identified entity pairs: "Is $e_i$ a contributing factor of $e_j$?" The LLM is prompted to respond "Yes" if a causal link exists or "No" otherwise. Since each direct or indirect cause typically contributes to a single node in the accident timeline, "Trigger" relationship extraction requires a broader contextual understanding of the entire accident chain. For this, question set B is used: "Which node in the accident timeline will $e_i$ result in thus contributing to the accident?"

By focusing on specific causal questions between identified entities, LLMs can reliably extract meaningful triples, unlike when all entities are listed indiscriminately. However, during repeated experiments, we observed that LLMs' responses to the same questions were not always consistent. To improve the reliability of extracting complex causal relationships related to construction accidents, this study introduces a self-consistency (SC) strategy. Fig. 5 illustrates this approach. The method first provides a set of diverse reasoning paths, where each reasoning path may lead to a different final answer. By randomly sampling multiple reasoning paths, each representing a separate CoT process, LLMs generate answers for each path. The final answer is determined by aggregating the most frequent answer from all reasoning paths. This approach enhances LLMs' ability to handle complex and variable relationship structures, as it simulates the human approach of cross-validating answers through diverse reasoning strategies. If multiple independent reasoning paths converge on the same answer, confidence in the correctness of that answer increases. The SC strategy reduces the randomness inherent in single-sample generation and provides stronger performance stability and generalization in extracting causal relationships. The prompts used in this method are shown in Fig. 6, where Task

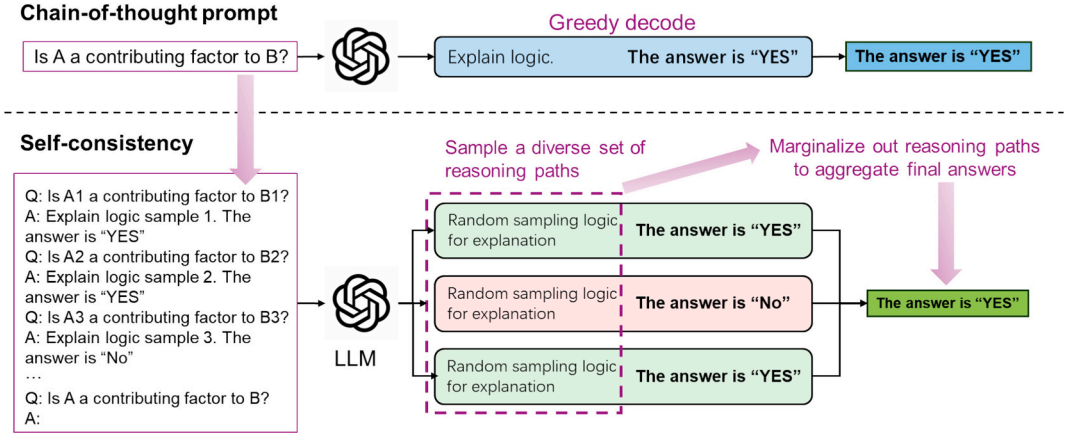



**Fig. 5.** Overview of the self-consistency strategy.

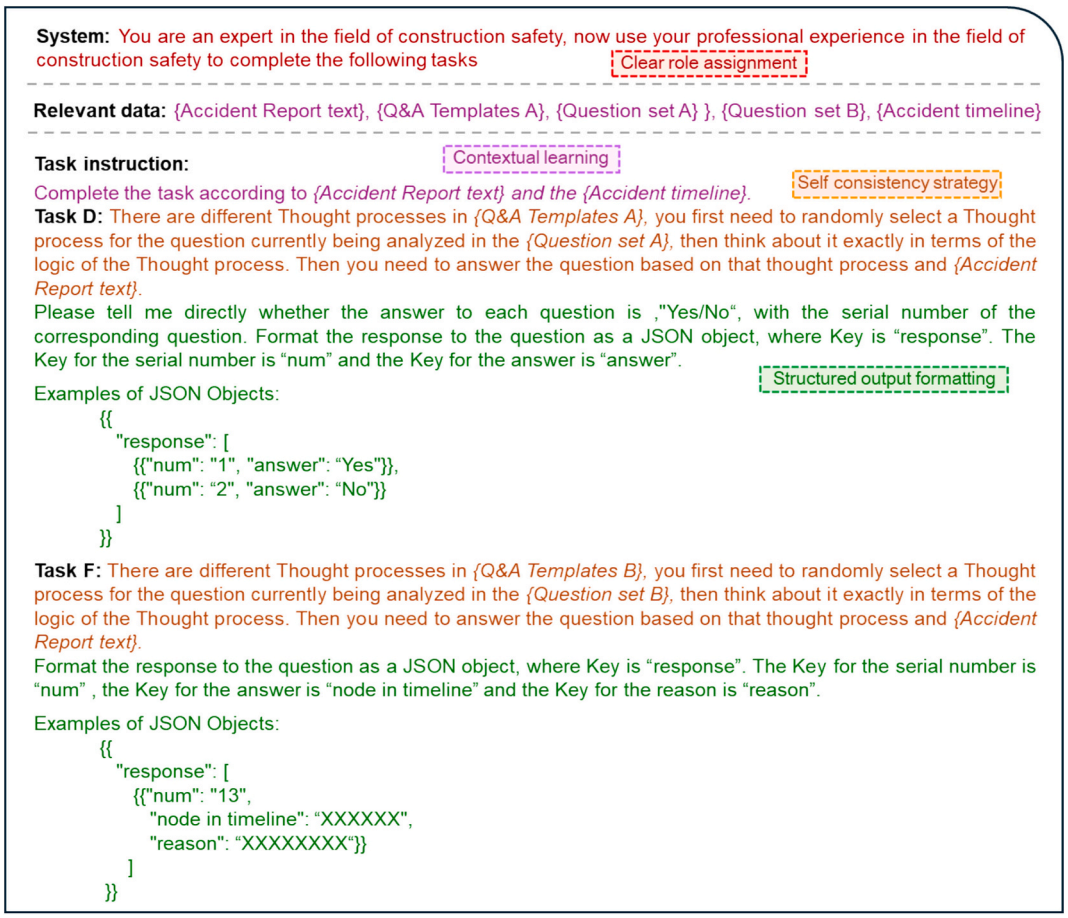



**Fig. 6.** Prompt structure of the RE task.

D is designed to extract "Cause" relationships, and Task F focuses on extracting "Trigger" relationships. In line with the SC strategy, each prompt needs to be executed multiple times, and the final relationship is selected via majority voting, thereby enhancing the robustness of the extracted relationship.

### 3.2.3. Evaluation of extraction performance

Assessing the accuracy of entities and relationships extracted by LLMs is essential to ensure the reliability and practical value of the FFHKG. Human expertise plays a crucial role in evaluating the quality and correctness of the extracted information and in identifying relevant entities that may have been missed. Therefore, manual annotation by domain experts is employed as the reference standard. For entity evaluation, experts first determine whether the extracted entity is mentioned in the report and whether its type is correctly identified. They also annotate any relevant entities present in the report that were not extracted by the model. For relationships, experts assess whether the linked entities are accurate and whether the relation is appropriately expressed in the report. To minimize subjective bias, multiple experts independently evaluate each extraction. Inter-annotator agreement is quantified using Fleiss' Kappa, which statistically measures the consistency among annotators and ensures the reliability of the reference standard [46]. Only the annotations that meet the agreement threshold are used as the final reference standard for subsequent performance evaluation.

This study employs well-known metrics to assess the quality of the extracted entities and relationships. The extraction quality of both entities and entity relationships will be evaluated separately. We first use the classic confusion matrix entries to evaluate the performance of extracting each component. Specifically, correctly extracted components are considered True Positives (TP), while incorrect ones are labeled as False Positives (FP), and missing entities are treated as False Negatives (FN). Based on these classifications, Precision (P) is computed as the ratio of correct results to all retrieved results, and Recall (R) is the ratio of correct results to all relevant results. The F1 score, defined as the harmonic mean of Precision and Recall, provides a balanced evaluation of both metrics.

For entity extraction, the following formulas are used:

$$P^E = \frac{1}{n}\sum_{i=1}^{n}\frac{TP_i^E}{TP_i^E + FP_i^E} \tag{1}$$

$$R^E = \frac{1}{n}\sum_{i=1}^{n}\frac{TP_i^E}{TP_i^E + FN_i^E} \tag{2}$$

$$F_1^E = \frac{2\cdot P^E \cdot R^E}{P^E + R^E} \tag{3}$$

where $n$ is the number of evaluators; $P^E$ represents the Precision of NER; $TP_i^E$ represents the count of correctly identified entities according to evaluator $i$; $FP_i^E$ indicates the count of incorrectly extracted entities according to evaluator $i$; $R^E$ represents the Recall of NER; $FN_i^E$ denotes the count of relevant entities missed according to evaluator $i$; $F_1^E$ represents the F1 score of NER.

Since the extraction of relationships is based on specific questions, there are no FN for relationships. Therefore, the evaluation metric for relation extraction is $P^R$. For relation extraction, the evaluation metric is:

$$P^R = \frac{1}{n}\sum_{i=1}^{n}\frac{TP_i^R}{TP_i^R + FP_i^R} \tag{4}$$

where $P^R$ represents the Precision of RE; $TP_i^R$ is the count of correctly extracted relationships according to evaluator $i$; $FP_i^R$ denotes the count of incorrectly extracted relationships according to evaluator $i$.

#### 3.2.4. Construction and visualization of FFHKG

After extracting the triples from the FFH accident reports, the next step is to map these triples to the FFHKG network. In the process of building the KG, nodes represent entities, and edges represent the relationships between those entities. Since causal relationships typically have a certain direction, the edges in the FFHKG are depicted as arrows rather than simple lines.

To represent the FFHKG, we define an adjacency matrix ($AM_{ij}$), which is a square matrix. $AM_{ij}$ is defined as follows:

$$AM_{ij} = \begin{cases} 1, \text{if } e_i \text{ and } e_j \text{ are connected by } r \\ 0, \text{otherwise} \end{cases} \tag{5}$$

where $AM_{ij} = 1$ indicates that a relationship exists from $e_i$ to $e_j$, while a value of 0 indicates no direct connection.

For the visualization of FFHKG, Neo4j is employed due to its user-friendly and interactive graphical interface, specifically optimized for graph database representations. Neo4j enables direct querying and dynamic rendering of KG using Cypher, a robust query language specifically designed for graph data.

### 3.3. Knowledge fusion of FFHKG

Accident reports are unstructured texts, and the expressions of various entities may be inconsistent. For instance, entities such as "insufficient safety training," "lack of full-time safety training," and "no safety training provided" all refer to the same concept of "insufficient safety training" but are expressed differently. Displaying all entities with the same meaning would significantly increase the complexity of the network. Therefore, it is necessary to fuse knowledge by merging multiple entities with the same meaning into a single entity. In this section, we first use pre-trained models for entity embedding, then apply unsupervised hierarchical clustering to return subsets of semantically similar entities or relationships, thus integrating the KG into a more understandable weighted complex network structure. Since the complex network is designed for quantitative analysis of influencing factors, basic information entities are excluded from the clustering process.

#### 3.3.1. Knowledge embedding

Knowledge embedding involves transforming textual information into vectorized numerical forms that preserve semantic relationships among words, sentences, or entire documents [47]. Based on textual granularity, embedding techniques are typically classified into word-level, sentence-level, and document-level embeddings. Since the knowledge entities obtained in the previous section are in the form of phrases, this study selects sentence-level embedding using pre-trained models, which map the entire phrase into fixed-length vectors.

However, sentence vectors generated by transformer-based models such as BERT are often constrained to a small region in the vector space and tend to exhibit excessive similarity, which limits their effectiveness in distinguishing semantically different entities. To address this issue, the Cosine Sentence (CoSENT) method was proposed to improve sentence representation by better aligning it with the data distributions of downstream tasks [48]. Based on CoSENT, a "shibing624/text2vec-base-chinese-paraphrase" model is trained, incorporating a contrastive loss function that maximizes the similarity of positive sentence pairs while minimizing the similarity of negative pairs [49], as defined in Eq. (6).

$$L = \log\left(1 + \sum_{\cos(x^i, x^{i+}) > \cos(x^i, x^{i-})} e^{\left(\cos(x^i, x^{i-}) - \cos(x^i, x^{i+})\right)/t}\right) \tag{6}$$

where $x^i$ denotes sentence vector of Chinese sentences; $(x^i, x^{i+})$ and $(x^i, x^{i+})$ represent the positive sample pair and negative sample pair in the semantic textual similarity training dataset, respectively; $t$ is the temperature coefficient.

This model achieved state-of-the-art (SOTA) performance on the Chinese Natural-Language Understanding test set, demonstrating its effectiveness in semantic matching tasks [49]. Therefore, it is selected in this study to vectorize knowledge entities.

#### 3.3.2. Knowledge clustering

After vectorizing the entities using pre-trained models, similar entities need to be aggregated through clustering. A common method for clustering is by the comparing cosine similarity between vectors. However, text similarity is more suitable for pairwise comparison and does not directly apply to generating different groupings. Clustering algorithms, a category of unsupervised learning techniques, divide datasets into groups of instances sharing similar characteristics. Common approaches include hierarchical clustering, K-means, and DBSCAN. In this study, however, the optimal number of clusters for each entity type is unknown, and cluster centroids cannot be predefined. Therefore, we use Hierarchical Agglomerative Clustering (HAC) to eliminate the need for initial group center selection and enhance the stability of the clustering results. HAC is a bottom-up single-linkage hierarchical clustering method [50]. It offers advantages in complex clustering scenarios, as it does not require predefining the number of clusters or initializing centroids and is less prone to local optima.

HAC starts by considering each data point as a separate cluster and iteratively merges the pair of clusters with the highest similarity until a specified stopping condition is reached. After the clustering process is complete, hierarchical results at any level can be obtained, providing extensive data to evaluate the similarity between factors, as shown in Fig. 7. During clustering, the feature vector sequence of entities is denoted as $F = \{f_1, f_2, \cdots, f_N\}$, where $N$ denotes the total number of entities. These vectors are partitioned by a set of segment boundaries $T = \{t_1, t_2, \cdots, t_S\}$, where $S$ denotes the total number of segments. The objective of HAC is to determine the optimal boundary set that minimizes the within-cluster square error function $f(F, T)$, as defined in Eq. (7). The number of segments $S$ corresponds to the final number of clusters $L$ when $f(F, T)$ reaches its minimum value. HAC allows for data-driven segmentation of entities to balance over-segmentation and under-segmentation without requiring manually predefined cluster numbers.

$$\min_T f(F, T) = \sum_{s=1}^{L} \sum_{t=t_{s-1}+1}^{t_s} \left\| f_t - \frac{1}{t_s - t_{s-1}} \sum_{t=t_{s-1}+1}^{t_s} f_t \right\|^2 \tag{7}$$

After aggregating semantically similar entities through hierarchical clustering, each cluster is assigned a representative category name, referred to as a risk factor. The final step involves mapping these clustered risk factors and their weighted causal links onto a weighted complex network. The weighted complex network is represented by the weighted adjacency matrix $AM^w$, defined by Eq. (8).

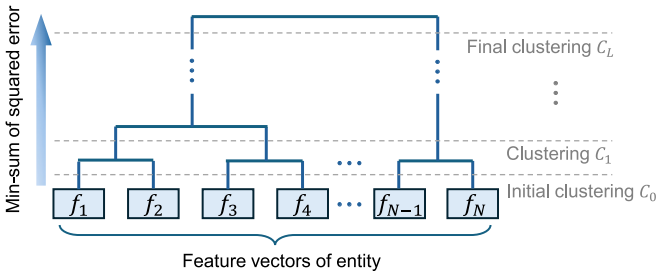



**Fig. 7.** Schematic diagram of the HAC principle.

$$AM_{ij}^{w} = \begin{cases} w_{ij}, \text{if } c_i \text{ and } c_j \text{ are connected by } r \\ 0, \text{otherwise} \end{cases} \tag{8}$$

where $w_{ij}$ represents the weight of the relationship from risk factor $c_i$ to $c_j$, determined by the frequency of the corresponding relationship $r$ in the knowledge triples. This matrix representation enables the transformation of the FFHKG into a structured complex network, visually capturing the weighted causal relationships among risk factors.

### 3.4. Topology analysis of the weighted complex network

After transforming the FFHKG into a weighted complex network, topology analysis offers a comprehensive perspective for accident analysis, helping to explore potential characteristics through network topology metrics. However, conventional topology metrics are designed for one-dimensional networks with similar types of links and nodes, making them unsuitable for analyzing the multi-dimensional weighted complex network. Therefore, novel metrics have been defined in this section to quantify risk factors.

**The degree is the first indicator used to characterize the direct relational structure of each** risk factor **in the** complex **network**. We introduce the concept of "out degree" (OD) to represent the number of factors that a factor $h$ can directly influence, denoted as $D_h^O$. Similarly, the "in degree" (ID) refers to the number of factors that directly influence $h$, denoted as $D_h^I$. A higher OD is structurally more influential, as it can directly affect a larger number of other factors. Conversely, A higher ID is more vulnerable, as it is more susceptible to being directly affected by other factors. The OD and ID of risk factor $h$ are calculated as follows:

$$D_h^O = \sum_{i \in HN} A_{hi} \tag{9}$$

$$D_h^I = \sum_{i \in HN} A_{ih} \tag{10}$$

where $A_{hi} = 1$ if factor $h$ has a direct influence on factor $i$, otherwise $A_{hi} = 0$; $HN$ represented the factor nodes other than $h$ in the complex network.

Causal proximity is the second indicator. We define "active causal proximity" (ACP) and "passive causal proximity" (PCP) to quantify the difficulty for factor $h$ to cause other factors, either directly or indirectly, and the difficulty for factor $h$ to be caused by the occurrence of other factors. These are denoted as $C_h^A$ and $C_h^P$, respectively. The greater the active causal proximity exhibited by a given risk factor, the higher the probability that said factor will cause other factors. Conversely, a factor with a higher passive causal proximity is more likely to be caused by other factors. Since a higher weight indicates a stronger causal relationship, the shortest path calculation is adjusted through a linear transformation of the edge weights. The transformed weight $w'_{jk}$ of edge $(j,k)$ and the causal proximity $C_h^A$and $C_h^P$ based on the adjusted shortest paths are expressed as follows:

$$w'_{jk} = w_{max} + w_{min} - w_{jk} \tag{11}$$

$$C_h^A = 1 \Big/ \left( \sum_{i \in HN} \frac{\sum_{(j,k) \in S(h,i)} w'_{jk}}{\sum_{i \in HN} u_{hi}} \right) \tag{12}$$

$$C_h^P = 1 \Big/ \left( \sum_{i \in HN} \frac{\sum_{(j,k) \in S(i,h)} w'_{jk}}{\sum_{i \in HN} u_{ih}} \right) \tag{13}$$

where $w_{jk}$ represents the original causal weight of edge$(j,k)$; $w_{max}$ and $w_{min}$ are the maximum and minimum weights in the network, respectively; $S(h,i)$ denotes the set of edges along the shortest path from factor $h$ to factor $i$; $u_{hi}$ denotes whether a shortest path exists between factor $h$ and $i$, taking the value of 1 if a path exists and 0 otherwise. By applying the adjusted shortest path calculation, the model effectively captures the role of strong causal relationships in hazard propagation.

To develop effective preventive strategies, the topology analysis also incorporates the assessment of risk value. This provides a decision-making basis for allocating resources to accident prevention efforts. The risk value of factor $h$, denoted as $R_h$, is computed as follows:

$$\mathrm{R}_h = normalization\left( \beta_h \cdot \left( \sum_{a \in EN} u_{ha} \cdot P_{(ah)} \cdot E_a \right) \right) \tag{14}$$

where $\beta_h$ represents the occurrence frequency of factor $h$; $E_a$ denotes the severity weight of the accident, which is assigned based on its economic loss. $P_{(a|h)}$ represents the probability that risk factor $h$ leads to accidents with economic loss $a$. $u_{ha}$ indicates whether a connection exists between factor $h$ and an accident with economic loss $a$, taking a value of 1 if a path exists; otherwise, 0.

It is important to note that factor $h$ can also act as an intermediary node in accident propagation within the complex network. To quantify the mediating role of each factor, we define "mediating centrality" (MC) as follows:

$$MC_h = normalization\left( \sum_{i \in HN, a \in EN} (u_{ih} \cdot u_{ha} \cdot R_i) \right) \tag{15}$$

where $u_{ih}$ indicates whether there is a connection between factor $i$ and factor $h$; $R_i$ denotes the risk value of factor $i$.

## 4. Case study

To validate the effectiveness and applicability of the proposed methodology, this section presents a case study based on real-world FFH accident data. A large-scale dataset of construction site FFH accident reports was collected and analyzed. The FFHKG was automatically constructed using LLMs and subsequently refined into a complex network to enable quantitative analysis of risk factors.

### 4.1. Data collection

To construct the database for construction site FFH accidents, this study collected a total of 4590 accident cases using web crawlers. The detailed accident reports were obtained from the "Safety Production Management" section on the official websites of various provincial and municipal governments in China. Additionally, using search engines on government websites, keywords such as "fall-from-height accidents," "accident reports," and "accident investigations" were employed to search for accident cases directly. After thorough filtering, 1097 accident cases were retained for the dataset in this study. The filtering process followed two main principles: (1) Only accidents with detailed reports, including direct and indirect causes, were retained. Many cases were recorded on websites with minimal information, making it difficult to extract risk factors. These cases were excluded from the dataset. (2) Only accidents in the past ten years were selected. Earlier data may not accurately reflect the current state of safety management on construction sites, as it may lack recent updates and safety practices. Therefore, accidents from 2014 to 2023 were chosen. Additionally, many accident reports are removed from government websites after some time, so the reports collected in this study represent all available reports from 2014 to 2023. While the 1097 cases are not the total number of FFH accidents in China during this period, they contain detailed information useful for further analysis. All reports were converted into .txt format for uniformity in data processing.

### 4.2. FHHKG construction and fusion

Building on the collected accident report dataset, this section

demonstrates the practical implementation of FFHKG construction and fusion. It presents the LLM-based extraction results and subsequent knowledge fusion process, which collectively support the development of a complex network for risk analysis.

#### 4.2.1. Extraction results of FFHKG

In this study, a prompt-based approach was implemented using LLMs. Using the constructed prompts, knowledge entities and relationships can be sequentially identified. Different LLMs exhibit varying performance in causal reasoning, long-text comprehension, and computational efficiency. To identify the most suitable LLM for knowledge extraction, we conducted a comparative evaluation of six leading LLMs.

We selected GPT-4o, Gemini 2.0 Flash, and Claude 3.7 Sonnet as representatives of closed-source LLMs. These closed-source models exhibit strong overall performance due to high-quality pretraining data and carefully optimized architectures. GPT-4o is known for its robust generalization and ability to reason in context. Claude 3.7 Sonnet, focusing on interpretability and instruction alignment, performs stably in structured extraction tasks. Gemini 2.0 Flash offers exceptional speed and cost efficiency, making it attractive for large-scale batch processing. In contrast, open-source models offer greater flexibility, transparency, and local deployment potential. For this category, we included Llama 3.3, Qwen 2.5, and DeepSeek R1. Qwen 2.5 is optimized for Chinese content understanding, making it particularly suitable for Chinese-language accident reports. DeepSeek R1, on the other hand, represents a model with strong reasoning abilities, especially in multi-hop tasks. Three experts were invited to manually annotate a random sample of 100 accident reports to assess the extraction accuracy. The inter-annotator agreement was evaluated using Fleiss' Kappa, resulting in a coefficient of 0.927, which indicates almost perfect agreement. This high consistency validates the reliability of the expert-annotated reference standard used for performance evaluation. The final evaluation metrics were calculated based on the average value of the three experts. Results are reported in Table 2.

Among all models, GPT-4o achieved the best overall performance. In the NER task, GPT-4o achieved the highest $F_1^E$ score of 0.94, with a precision $P^E$ of 0.97, which is also the highest across all models. Claude 3.7 Sonnet ranked second with $F_1^E = 0.92$ but incurred a higher cost and longer processing time. Gemini 2.0 Flash was the most cost-effective and responsive among the proprietary models but lagged behind in both precision and recall. Among open-source models, Qwen 2.5 exhibited the strongest $R^E$ of 0.93, benefiting from its enhanced Chinese pretraining corpus and alignment with domain-specific syntax. DeepSeek R1 delivered the highest $P^E$ of 0.94 within the open-source group, although its recall was limited ($R^E = 0.86$), possibly due to loss of context in complex reasoning chains. LLaMA 3.3 showed the weakest NER performance among all models, with $F_1^E$ of only 0.74. In the RE task, GPT-4o again outperformed all other models and was the only one to achieve a $P^R$ of 0.90. DeepSeek R1 also performed well ($P^R = 0.89$), likely due to its strong reasoning capabilities, but at the cost of significant latency, requiring an average of 351.03 s per case. The remaining models showed moderate and comparable performance in the RE task. In conclusion, ChatGPT-4o has advantages in speed, cost, and text reasoning capabilities. Given the zero-shot nature of the task and the inherent complexity of accident report texts, these results are highly encouraging. The integration of CP, CoT, and SC plays a crucial role in mitigating hallucination by guiding the model through structured reasoning and consistent answer generation.

**Table 2**
Comparison of evaluation metrics between six different LLMs.

| Models (FS & CoT & SC) | $P^E$ | $R^E$ | $F_1^E$ | $P^R$ | Cost | Time |
|---|---|---|---|---|---|---|
| ChatGPT-4o | 0.97 | 0.92 | 0.94 | 0.90 | 0.15 USD | 79.5 s |
| Gemini 2.0 Flash | 0.88 | 0.89 | 0.89 | 0.84 | 0.04 USD | 45.1 s |
| Claude 3.7 Sonnet | 0.92 | 0.92 | 0.92 | 0.83 | 0.23 USD | 106.5 s |
| Llama 3.3 | 0.75 | 0.72 | 0.74 | 0.83 | 0 USD | 298.4 s |
| Qwen 2.5 | 0.90 | 0.93 | 0.91 | 0.84 | 0 USD | 151.7 s |
| DeepSeek R1 | 0.94 | 0.86 | 0.90 | 0.89 | 0 USD | 351.0 s |

Therefore, ChatGPT-4o was chosen as the information extraction model, and a Prompt-based approach was implemented using the OpenAI API via Python. Using the constructed prompts, knowledge entities and relationships were sequentially identified from the 1097 collected fall-from-height accident reports. The resulting FFHKG contains a comprehensive causal network with 14,466 nodes and 49,788 edges. Due to the large size of the FFHKG constructed from the 1097 reports, a partial FFHKG consisting of the first 40 reports is shown in Fig. 8 for better visualization.

#### 4.2.2. Consistency of knowledge

To integrate semantically similar knowledge entities, entity texts were first vectorized using the text2vec-base-chinese-paraphrase model from the Text2Vec toolkit, which encodes each entity into a 768-dimensional dense vector space. HAC was then applied separately to eight categories of causal-related entities to automatically group those with similar semantic meanings. After clustering, each entity category yielded a distinct number of clusters, as shown in Table 3, with each cluster representing a unified risk factor. For example, in the "Con" category, several entities such as "inadequate safety training by contractors", "inadequate training for temporary workers", "insufficient training for employees", among others, were successfully grouped into the same cluster due to their shared semantics and were treated as one consolidated risk factor.

To facilitate visualization of the clustering results, the 768-dimensional vectors were reduced to two dimensions using Principal Component Analysis (PCA). The clustering outcomes for all eight causal-related entity categories are presented in Fig. 9. It can be observed that the entity clusters exhibit clear boundaries and distinct distributions, indicating that the embedding and clustering process effectively capture semantic similarities among the entities. To further evaluate the reasonableness of the hierarchical clustering structure, we selected the "Own" category and reduced its vector representations to six dimensions using PCA for visualization. The resulting dendrogram is shown in Fig. 10, where only three clusters are partially displayed due to the large number of nodes. We can see that entities within each cluster display similar vector patterns across principal components, confirming the internal semantic consistency of the clustered groups.

After obtaining the clustering results, we reviewed the semantic content and keyword patterns within each cluster to define representative risk factor names, as shown in Table A1 in Appendix A. In the FFHKG, the two entities in each triple were replaced with their corresponding category names, which became the nodes in the complex network, while the relationship remained the edge attribute. Using Gephi's powerful data manipulation and graph construction capabilities, we imported the standardized entity nodes and relationships in Excel format, automatically generating the graph. Fig. 11 reveals the causal relationship-based visualization of the weighted complex network, where node sizes indicate the sum of the weights of their connected edges, and edge width indicates path weight. Node corresponds to the standardized categories in Appendix A. This weighted representation enhances the weighted complex network by quantifying the strength of causal influences, enabling a more accurate depiction of hazard propagation and accident risk assessment.

### 4.3. Quantitative analysis of risk factors

To quantitatively assess the risk factors of FFH accidents, we perform a topology analysis of the weighted complex network derived from the clustering process. Based on the topology analysis indicators introduced in Section 3.4, we explore the structural properties and key influencing factors. Out degree and in degree of risk factors are presented in Fig. 12.

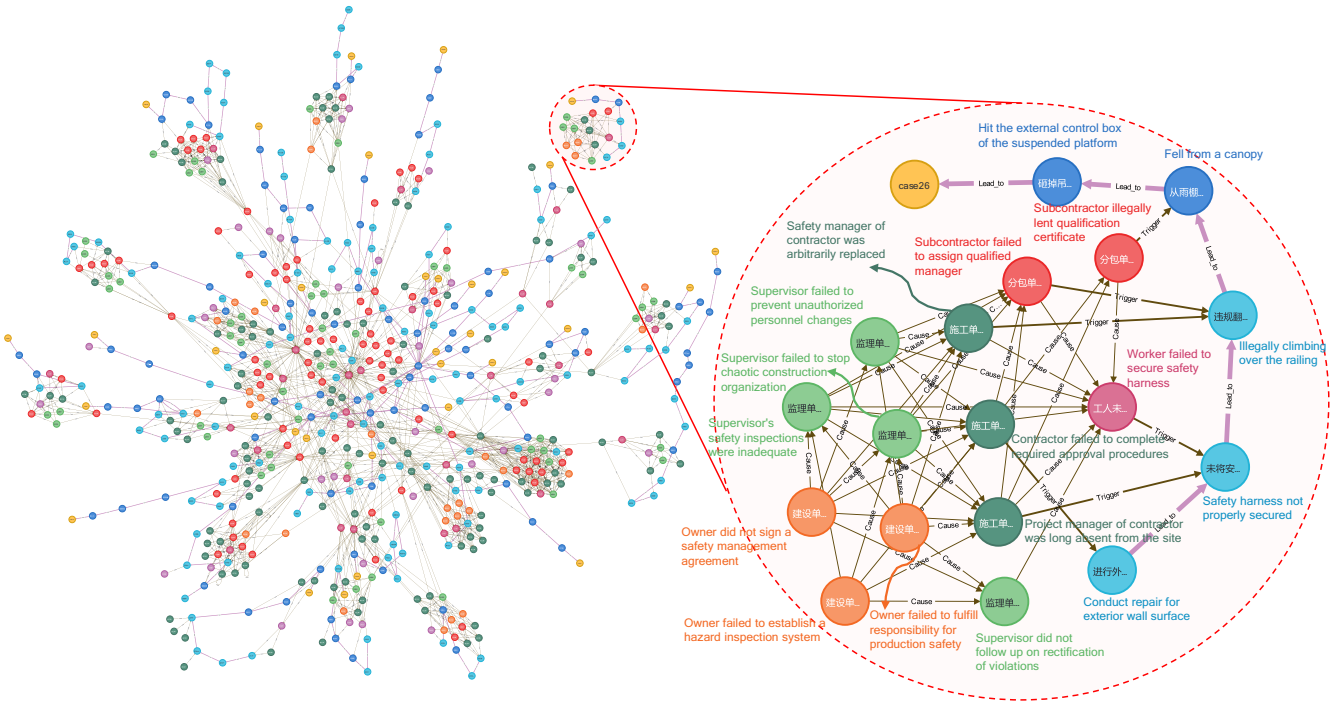



**Fig. 8.** Full and partial view of FFHKG extracted from 40 accident reports.

**Table 3**
Distribution of nodes and clusters in different entity categories.

| Categories | Number of entities | Number of clusters | Proportion of total entities |
|---|---|---|---|
| Own | 775 | 42 | 6 % |
| Sup | 829 | 40 | 6 % |
| Con | 3078 | 56 | 24 % |
| Sub | 1452 | 40 | 11 % |
| Equ | 523 | 21 | 4 % |
| Surr | 948 | 38 | 7 % |
| Act | 3091 | 65 | 24 % |
| Sta | 2243 | 13 | 17 % |
| Total | 12,939 | 315 | 100 % |

Active causal proximity and passive causal proximity of risk factors are shown in Fig. 13. Risk value and mediating centrality for risk factors of risk factors is depicted in Fig. 14. Table 4 provides the statistical summary of out degree and in degree for each category. Table 5 shows the statistical summary of active causal proximity and passive causal proximity for each category. The quantitative results for each risk factor are presented in Table A2 in Appendix A. Detailed discussions of these results are provided below.

(1) Company management factors tend to exert broader direct influence and are more likely to trigger adverse outcomes through diverse pathways. In contrast, equipment-related issues and unsafe behaviors are typically more susceptible to upstream factors. For the OD, as shown in Table 4, the "Con" category shows the highest average OD at 56.4, indicating that contractor nodes are frequently connected to downstream operational risks. Their position enables them to act as hubs that translate management-level failures into concrete unsafe conditions on-site. In contrast, the "Surr" category exerts the lowest impact, with an average OD of only 7.7, suggesting that environmental hazards often exist as static conditions rather than dynamic propagation agents. Notably, as presented in Fig. 12(a), five factors exceed an OD value of 125, four of which are contractor-related: Con46 ("Inadequate on-site safety management"), Con51 ("Insufficient hazard investigation"), Con15 ("failure to fulfill safety responsibilities"), and Con5 ("Inadequate safety education"). Their exceptionally high connectivity highlights their central role in hazard propagation and makes them critical leverage points for prevention. The only non-Con factor exceeding this threshold is Sup21 ("Inadequate supervision"), which reflects that ineffective supervision can create widespread vulnerabilities, allowing issues across organizational levels to remain uncorrected. For the ID, as shown in Table 4, the "Sta" category has the highest average ID value of 80.2. This correctly reflects the multifactorial and systemic nature of accident occurrence in construction settings. Specifically, in Fig. 12(b), Sta10 ("Worker falling"), Equ1 ("Failure to wear safety harness"), and Sta8 ("Equipment/material instability") all exceeded the ID threshold of 125, making them the top-ranking nodes in passive connectivity. The dominance of Sta10 and Sta8 underscores the complexity of FFH accidents, which often result from a convergence of factors. Meanwhile, the high ID of Equ1 reveals that safety harness non-compliance is not merely an individual behavioral lapse but a manifestation of deeper systemic issues. In conclusion, the indicator provides a clear basis for identifying core initiators and highly vulnerable endpoints, thus offering targeted insights for optimizing safety interventions and enhancing the robustness of construction risk management systems.

(2) The distribution patterns of causal proximity across different factor categories align well with the structural logic of the KG ontology, supporting the reliability of the metric. Firstly, as presented in Fig. 13 (a), management-related nodes positioned at the upper level of KG exhibit both high ACP and PCP values due to their strong top-down influence and involvement in key causal chains. As shown in Table 5, the "Con" category has the highest average ACP of $2.547 \times 10^{-3}$, indicating that contractor-related issues are the most influential initiators. Notably, Con46 plays a pivotal role in triggering downstream events. In contrast, the "Sub" category shows the lowest ACP, with a value of $2.306 \times 10^{-3}$, reflecting subcontractors' limited authority in safety coordination and their function as task executors. SUP nodes have the

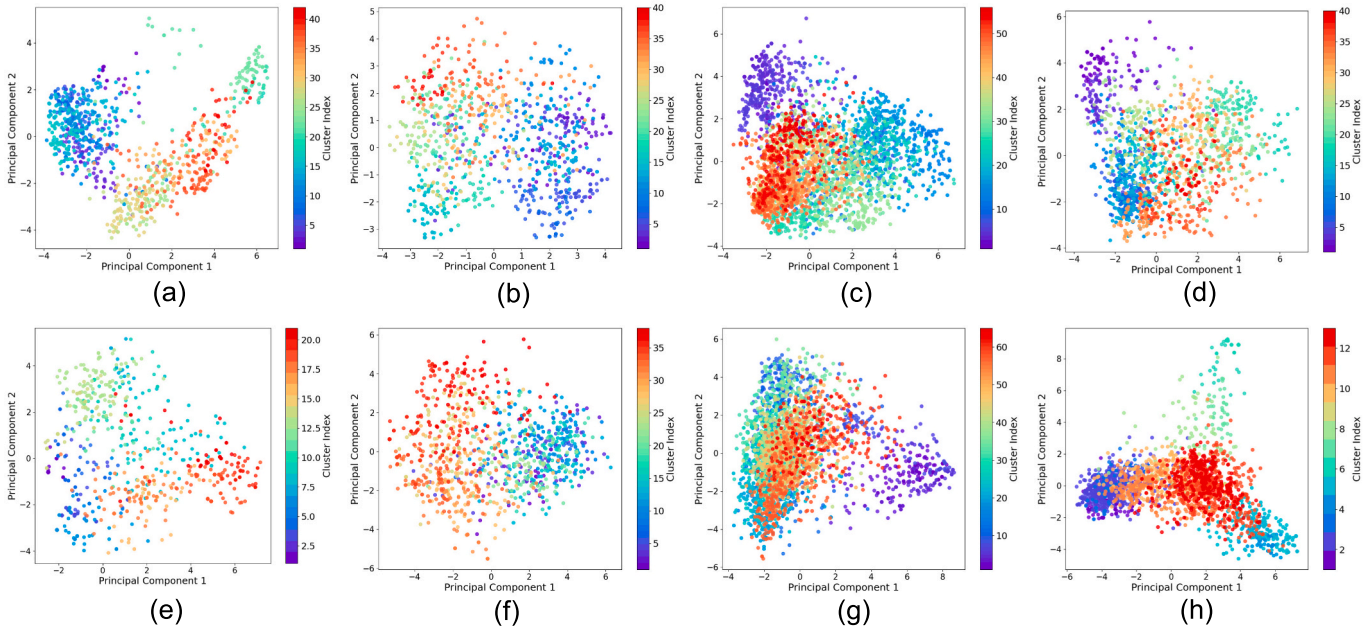



**Fig. 9.** Cluster scatter diagram of different entity categories: (a) Own; (b) Sup; (c) Con; (d) Sub; (e) Equ; (f) Surr; (g) Act; (h) Sta.

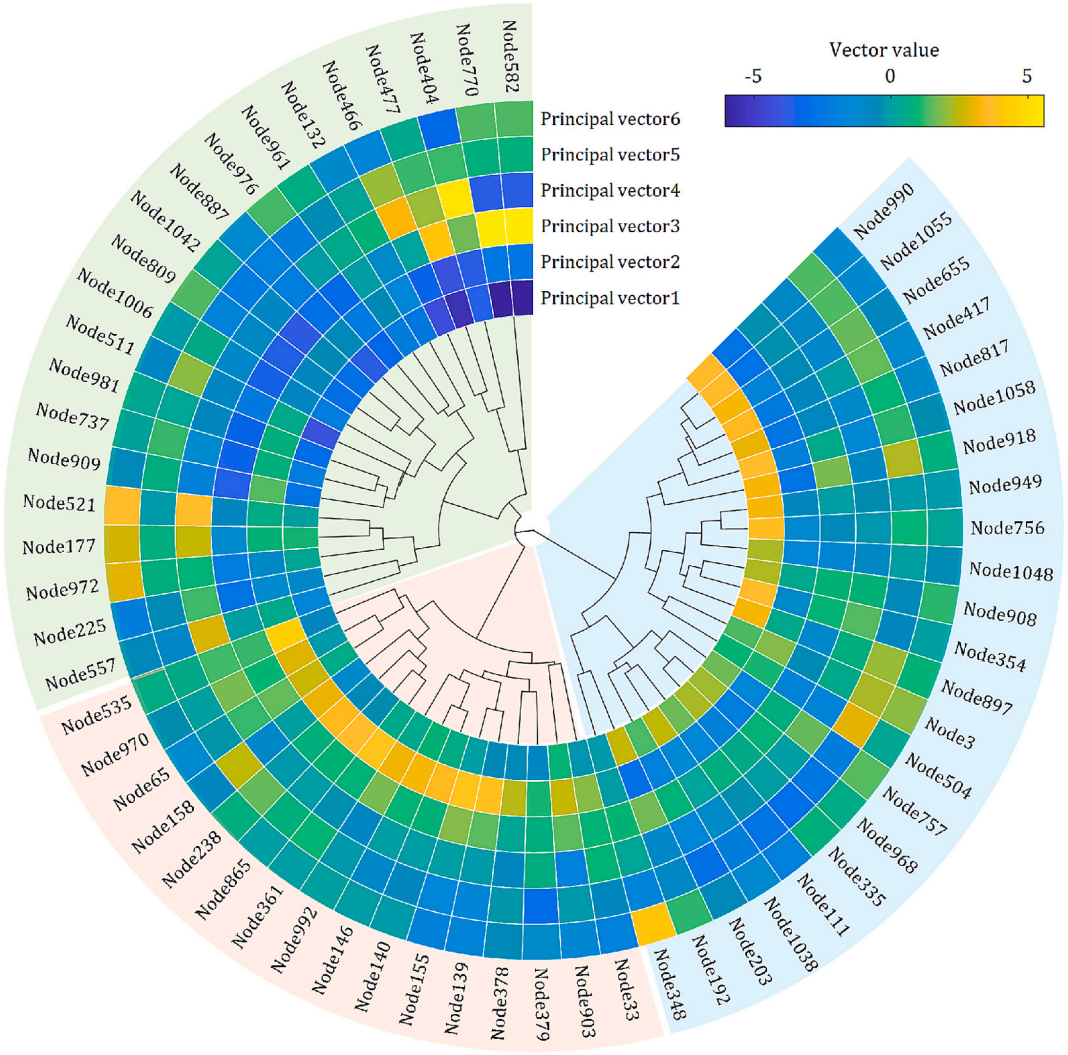



**Fig. 10.** Partial dendrogram of three clusters in the "Own" category.

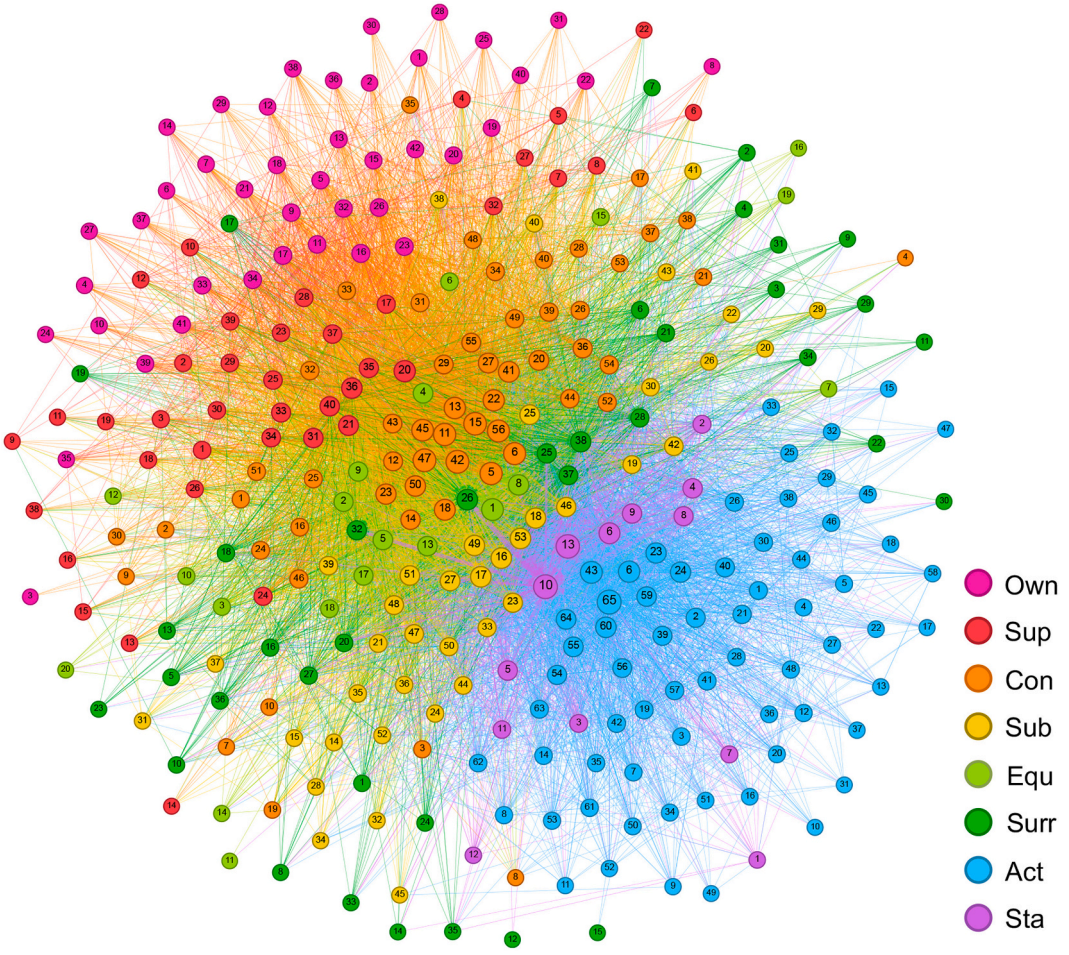



**Fig. 11.** Weighted complex network transferred from FFHKG.

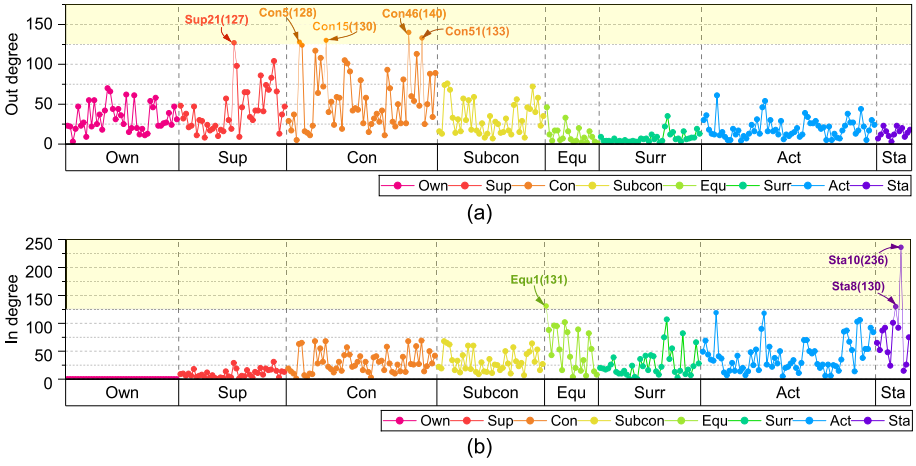



**Fig. 12.** Out degree and in degree of risk factors in different categories: (a) out degree; (b) in degree.

highest average PCP, reaching $4.654 \times 10^{-3}$, which can be attributed to their direct links from owner-related nodes with strong causal weights despite having fewer incoming paths. An exception is Sup15 ("Unclear responsibilities of supervision personnel"), with a PCP of 0, indicating it is modeled as an independent root-cause issue. Similarly, all Owner nodes have zero PCP, reinforcing their exclusive role as initiators rather than recipients in the network. Secondly, as presented in Fig. 13 (b), within physical factors, the "Equ" category exhibits higher ACP and PCP than the "Surr" category, suggesting that equipment-related issues are more central in initiating and receiving hazards. Equ1 and Surr26 ("No edge protection") exhibit the highest ACP values in their respective groups, indicating their dominant roles in initiating causal chains. Meanwhile, Equ8 ("Failure to wear safety helmet ") and Surr26 show the highest PCP values, suggesting that these nodes function as critical convergence

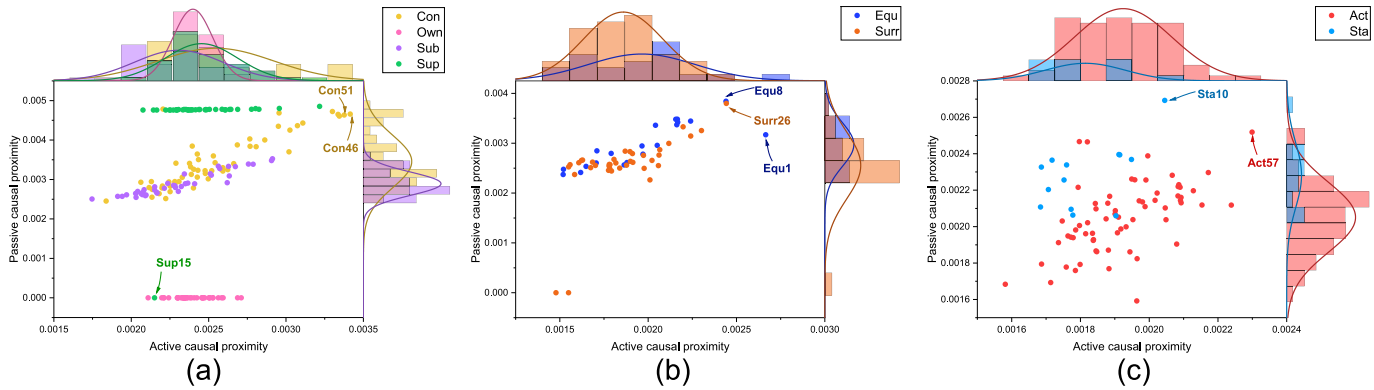



**Fig. 13.** Active causal proximity and passive causal proximity of risk factors in different categories: (a) management-related factors; (b) physical context factors; (c) event-sequence factors.

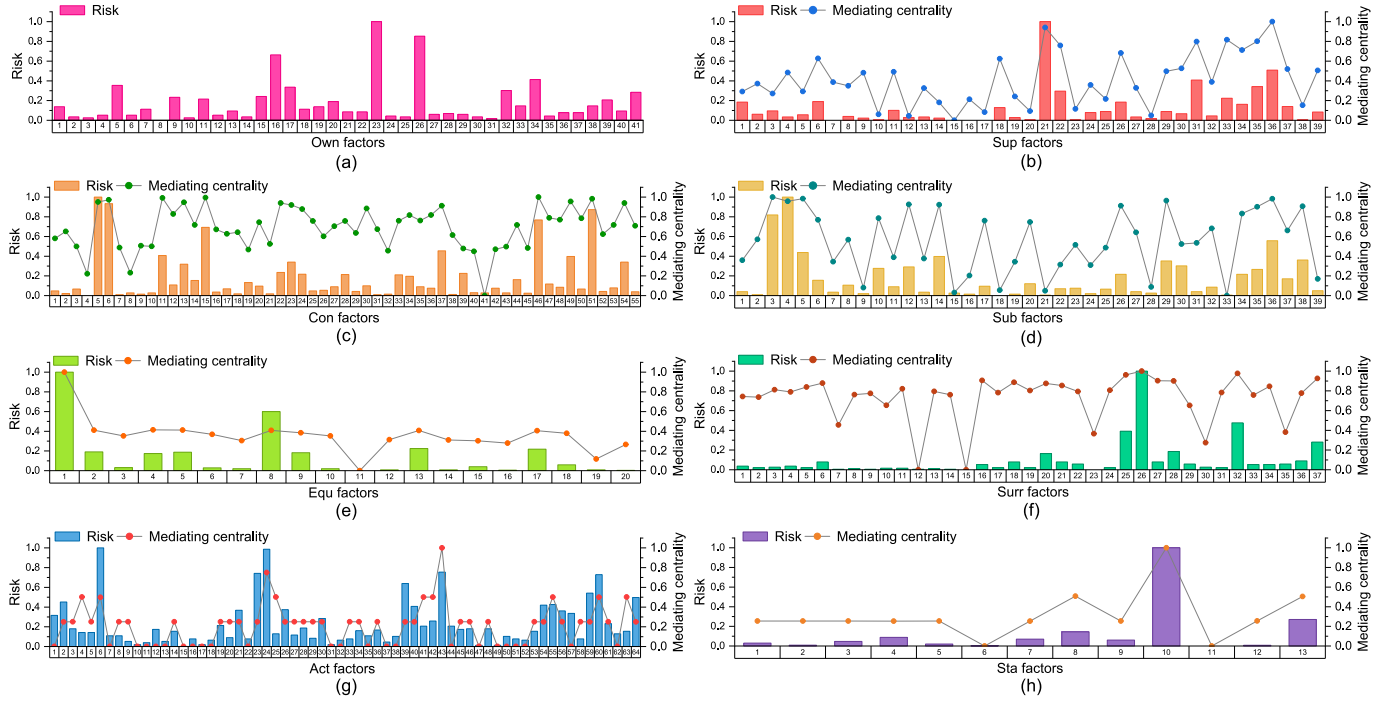



**Fig. 14.** Risk value and mediating centrality of risk factors in different categories: (a) Own; (b) Sup; (c) Con; (d) Sub; (e) Equ; (f) Surr; (g) Act; (h) Sta.

**Table 4**
Statistical summary of out degree and in degree for each category.

| | | Own | Sup | Con | Sub | Equ | Surr | Act | Sta |
|---|---|---|---|---|---|---|---|---|---|
| OD | Max value | 70 | 127 | 140 | 76 | 46 | 35 | 61 | 23 |
| | Max risk factor | Own16 | Sup21 | Con46 | Sub4 | Equ1 | Surr26 | Act6 | Sta8 |
| | Min value | 3 | 8 | 5 | 7 | 1 | 2 | 4 | 3 |
| | Min risk factor | Own3 | Sup10 | Con4 | Sub21 | Equ20 | Surr23 | Act15 | Sta6 |
| | Mean value | 32.7 | 42.4 | 56.4 | 32.4 | 11.5 | 7.7 | 19.7 | 14.1 |
| ID | Max value | 0 | 31 | 69 | 68 | 131 | 107 | 119 | 236 |
| | Max risk factor | N. A | Sup36 | Con15 | Sub3 | Equ1 | Surr26 | Act6 | Sta10 |
| | Min value | 0 | 0 | 1 | 7 | 1 | 0 | 5 | 15 |
| | Min risk factor | N. A | Sup15 | Con4 | Sub33 | Equ11 | Surr12, Surr15 | Act31 | Sta11 |
| | Mean value | 0 | 9.9 | 30.2 | 31.6 | 53.6 | 26.4 | 41.1 | 80.2 |

points where multiple upstream factors accumulate. Finally, as presented in Table 5, the "Act" and "Sta" categories rank lowest in both ACP and PCP, consistent with their roles as consequence-bound elements. Individual behaviors are reactive, and accident states typically conclude the causal chains. Nevertheless, as shown in Fig. 13 (c), Act57 ("Worker position shift") and Sta10 stand out with the highest ACP and PCP in their respective categories. Act57 functions as a transitional behavior linking human and environmental risks, while Sta10, which appears in every causal chain due to the study's focus, underscores its semantic

**Table 5**
Statistical summary of active causal proximity and passive causal proximity for each category.

| | | Own | Sup | Con | Sub | Equ | Surr | Act | Sta |
|---|---|---|---|---|---|---|---|---|---|
| ACP | Max value | 2.711 | 3.217 | 3.415 | 2.913 | 2.665 | 2.442 | 2.299 | 2.045 |
| $\times 10^{-3}$ | Max risk factor | Own23 | Sup21 | Con46 | Sub4 | Equ1 | Surr26 | Act57 | Sta10 |
| | Min value | 2.109 | 2.076 | 1.838 | 1.747 | 1.519 | 1.478 | 1.581 | 1.684 |
| | Min risk factor | Own21 | Sup10 | Con4 | Sub33 | Equ11 | Surr15 | Act63 | Sta5 |
| | $\mu$ | 2.398 | 2.454 | 2.547 | 2.306 | 1.964 | 1.859 | 1.924 | 1.814 |
| | $\sigma$ | 0.127 | 0.232 | 0.373 | 0.283 | 0.297 | 0.222 | 0.140 | 0.112 |
| PCP | Max value | 0.000 | 4.854 | 4.785 | 3.525 | 3.842 | 3.799 | 2.519 | 2.693 |
| $\times 10^{-3}$ | Max risk factor | N. A | Sup21 | Con41 | Sub4 | Equ8 | Surr26 | Act57 | Sta10 |
| | Min value | 0.000 | 0.000 | 2.454 | 2.505 | 2.163 | 0.000 | 1.592 | 2.059 |
| | Min risk factor | N. A | Sup15 | Con4 | Sub33 | Equ11 | Surr12, Surr15 | Act45 | Sta11 |
| | $\mu$ | 0.000 | 4.654 | 3.457 | 2.894 | 2.923 | 2.555 | 2.051 | 2.282 |
| | $\sigma$ | 0.000 | 0.745 | 0.660 | 0.292 | 0.460 | 0.666 | 0.191 | 0.172 |

and structural centrality in FFH accidents. In conclusion, the causal proximity analysis offers a data-driven foundation for identifying priority targets in construction safety interventions from a causal perspective.

(3) The evaluation of FFH accident risks highlights significant differences across factor categories. As shown in Fig. 14, in the "Own" category, Own23 ("Illegal subcontracting to unqualified companies or individuals") and Own26 ("Unauthorized construction without basic construction procedures") have the highest risk values, emphasizing the need for stricter contractor selection and project oversight. In the "Sup" category, Sup21 poses the greatest risk due to inadequate safety management, while Sup36 ("Ineffective enforcement of hazard rectifications") serves as a key transmission link, reflecting the impact of ineffective hazard rectification. Risk value and MC are relatively consistent in the "Con" and "Sub" categories, with Con5 and Sub4 (related to insufficient safety education), and Con6 and Sub3 (related to failure to conduct safety education) scoring high, underscoring the importance of safety education. Con46 and Sub3 have the highest MC, reinforcing the role of site management and training in accident transmission. In the "Equ" category, Equ1 and Equ8 show high-risk value and MC, requiring priority intervention. Similarly, in the "Surr" category, Surr26 ranks highest, highlighting the need for stronger edge protection. Among unsafe behaviors, Act6 ("Climbing") and Act24 ("Doing dismantling work") present the highest risk values, reflecting the dangers of climbing and dismantling at heights. While accident states generally show lower risk values, Sta10 ranks highest, emphasizing the severe consequences of falls once they occur. In conclusion, effective accident prevention requires a comprehensive safety management approach, addressing vulnerabilities across construction, supervision, and subcontracting units, equipment, environmental factors, and unsafe behaviors, focusing on high-risk elements and key transmission points.

## 5. Discussion

To further examine the effectiveness of the proposed methodology and its relevance to practical construction safety improvement, this section presents additional experimental validation and corresponding risk mitigation insights.

### 5.1. Ablation experiment

This study introduces an LLM-powered approach for extracting causal knowledge from accident reports without labeled data as a novel methodological attempt, achieving promising results. To verify the effectiveness of each component in our proposed prompt engineering, we conducted ablation experiments in this section.

Prompt Chain (PC), Chain of Thought (CoT), and Self-Consistency (SC) constitute the three core components of the prompt engineering strategy proposed in this study. To assess the individual and combined contributions of these components, we conducted an ablation experiment evaluating GPT-4o under five distinct learning scenarios: (1) employing all three techniques (PC, CoT, and SC), (2) removing CoT, (3) removing SC, (4) removing PC, and (5) removing all three components. The results are summarized in Table 6.

The findings clearly demonstrate the necessity of integrating all three components to maximize information extraction performance. When PC, CoT, and SC were jointly applied, GPT-4o achieved the highest scores across all evaluation metrics, confirming the effectiveness of this prompt design in enhancing both accuracy and consistency. The removal of CoT resulted in a significant performance drop. In the NER task, $F_1^E$ fell from 0.94 to 0.77, and the RE task $P^R$ decreased from 0.90 to 0.81. This highlights the critical role of CoT in guiding the model through structured reasoning steps, particularly in tasks that involve complex inference and logical decomposition. Excluding SC also led to a marked reduction in performance. $P^R$ declined significantly to 0.84, indicating that SC helps enhance relation extraction reliability by mitigating hallucinations through iterative sampling and majority voting. This is especially important for tasks requiring multi-step validation of entity relationships. Similarly, removing PC negatively impacted model performance, reinforcing the importance of task decomposition. In the direct triple extraction setting, named entity recognition becomes an implicit process, increasing the cognitive burden on the model. By breaking down complex prompts into well-defined subproblems, PC allows the model to address each step more effectively, reducing reasoning overhead while maintaining logical consistency. When all three components were removed, overall performance deteriorated substantially. The $F_1^E$ dropped to 0.69, and although the decline in $P^R$ was less drastic than in the NER task, it still reached the lowest value of 0.78. In conclusion, combining PC, CoT, and SC significantly enhances LLM performance by improving reasoning depth, reducing error propagation, and promoting extraction stability. These components work synergistically to optimize entity recognition and relation extraction in complex KG construction tasks.

### 5.2. Recommendations for mitigating risk factors for FFH accidents

Based on quantitative analysis from the perspectives of direct causation, indirect causation, and risk propagation, key risk factors have

**Table 6**
Performance comparisons of ablation experiments.

| Models | $P^E$ | $R^E$ | $F_1^E$ | $P^R$ |
|---|---|---|---|---|
| GPT-4o (PC & CoT & SC) | 0.97 | 0.92 | 0.94 | 0.90 |
| GPT-4o (without CoT) | 0.85 | 0.70 | 0.77 | 0.81 |
| GPT-4o (without SC) | N. A | N. A | N. A | 0.84 |
| GPT-4o (without PC) | 0.88 | 0.74 | 0.80 | 0.86 |
| GPT-4o (without PC & CoT & SC) | 0.72 | 0.67 | 0.69 | 0.78 |

been identified across three domains. Grounding the findings of this study, some key intervention measures are proposed.

In the organizational domain, inadequate safety management, insufficient hazard identification, and lack of safety training emerged as dominant contributors, aligning with existing findings [28,30,51,52]. Notably, the inadequate safety education of the contractor and subcontractor exhibits significantly higher risk values than other risk factors. Future projects should enhance the accessibility and effectiveness of safety education while fostering proactive worker participation, which is critical for sustained behavioral change. Although construction owners are not directly involved in site operations, their decisions significantly shape upstream risk structures. Strengthening safety awareness training for owners can create a top-down safety culture, influencing supervisors, contractors, and subcontractors.

In terms of equipment safety, failure to wear safety harnesses and helmets are the most critical risk factors, issues that have been widely discussed in existing academic research [53,54]. Regarding environmental safety, insufficient edge protection and inadequate hole protection are identified as primary hazards, consistent with the existing conclusions [55,56]. These findings validate the effectiveness of the methodology proposed in this study. Unlike traditional approaches that rely on expert judgment, this research derives data-driven insights from a large dataset of historical accident reports, ensuring the robustness of its conclusions. Frequent references to safety management deficiencies and inadequate hazard detection indicate that site supervision solely reliant on designated personnel is insufficient. Once unsafe behaviors occur, they may rapidly escalate into serious accidents. Therefore, safety monitoring should be enhanced through a combination of automated equipment inspection, environmental optimization, and AI-based violation detection to interrupt accident development paths.

Additionally, since accident reports primarily document observable unsafe behaviors, this study points to positional shifts, climbing activities, and demolition operations as key contributing factors. In contrast, some research places greater emphasis on workers' personal safety habits and risk perception [57,58], highlighting a difference in research focus that complements the findings of this study. By focusing on observable actions and accident status, the results of this study offer additional insights into actionable risk mitigation. Among all unsafe actions, climbing-related activities pose the highest risk. This suggests that future posture recognition systems for high-altitude operations should prioritize the detection of climbing motions to enable timely warnings and interventions. Finally, demolition work was found to carry the highest overall risk among task types, likely due to unstable structures and dynamic environments. Enhanced supervision and targeted protective measures are essential for workers performing demolition at heights.

## 6. Conclusions

This paper proposed an LLM-driven framework for automated KG construction and causal analysis of FFH accidents. By integrating a refined AcciMap-based ontology, prompt engineering strategies, and network-based topology metrics, the proposed method enables robust extraction of structured causal information from unstructured accident reports and supports comprehensive, data-driven risk factor quantitative evaluation.

This study finds the following significant findings: (1) GPT-4o achieved the best overall performance in both entity and relation extraction tasks, with an $F_1^E$ of 0.94 and $P^E$ of 0.97 in NER, and a $P^R$ of 0.90 in RE. (2) The prompt design combining CP, CoT, and SC significantly improved model performance. Removing these components led to a drop in NER $F_1^E$ to 0.69 and in RE $P^R$ to 0.78. (3) The proposed framework objectively quantifies key risk factors extracted from a large number of accident reports from three perspectives. (4) Management-related issues, like inadequate safety management and lack of safety training, exhibit higher OD, causal proximity, and MC. Their centrality in causal pathways underscores their pivotal role in accident initiation and propagation. (5) Equipment and environment-related risks, including failure to wear harnesses and helmets and insufficient edge protection, etc., demonstrated high ID, causal proximity, and risk values, indicating they are frequent convergence points of upstream causes. (6) Observable behaviors such as unsafe climbing, demolition operations, and positional shifts showed higher risk values among all activities, offering practical insights for behavior-focused safety monitoring systems.

Despite its promising results, this paper has several limitations. First, the model may fail to identify implicit or contextually embedded entities, especially those not explicitly stated in the text. Second, the current prompt strategy adopts a fixed reasoning structure, which may limit the model's ability to adapt to diverse accident scenarios. Third, while self-consistency improves stability, it relies on a relatively simple majority voting mechanism that may not fully capture nuanced causal inferences. Future research could explore adaptive prompting strategies that include probing questions or counterfactual reasoning to enhance the extraction of implicit risk factors. Incorporating iterative voting mechanisms and expanding reasoning diversity could further reduce hallucination and improve inference accuracy. Moreover, as this paper focuses on textual data, future work could integrate computer vision techniques to link textual risk factors with observable behaviors in video data, enabling more comprehensive and real-time safety monitoring on construction sites.

## CRediT authorship contribution statement

**Qiong Liu:** Visualization, Methodology, Data curation, Conceptualization, Writing – original draft. **Yuexiong Ding:** Visualization, Software, Data curation, Writing – review & editing. **Xiaowei Luo:** Supervision, Resources, Funding acquisition, Conceptualization, Writing – review & editing.

## Declaration of competing interest

We declare that the manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

## Acknowledgements

The Hong Kong Research Grant Council PJ# 11211622 and the City University of Hong Kong Internal Fund PJ#9680139 jointly supported this work. The conclusions herein are those of the authors and do not necessarily reflect the views of the sponsoring agencies.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.autcon.2025.106482.

## Data availability

Data will be made available on request.

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